

Lab 10

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CO1: Develop fundamentals in fabrication of MOSFETs and MOS transistor

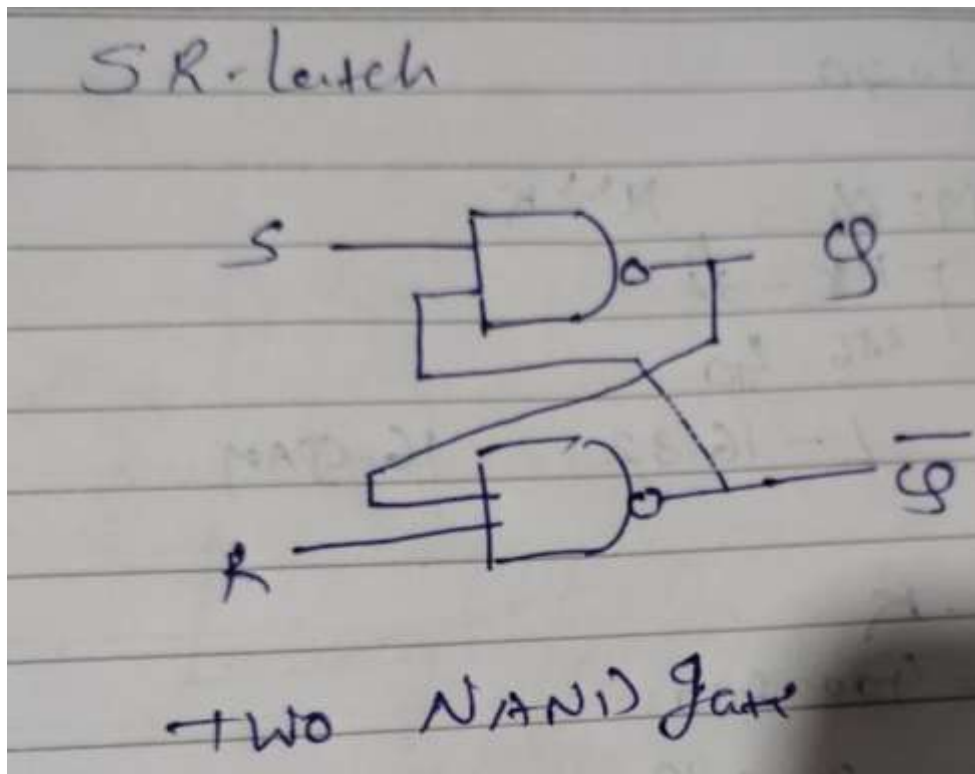
CO2: Analyze MOS inverters with its static and switching characteristics

CO3: Analyze Combinational and Sequential MOS logic circuit

Experiment 10: To draw the layout of SR-latch using CMOS and simulate the design

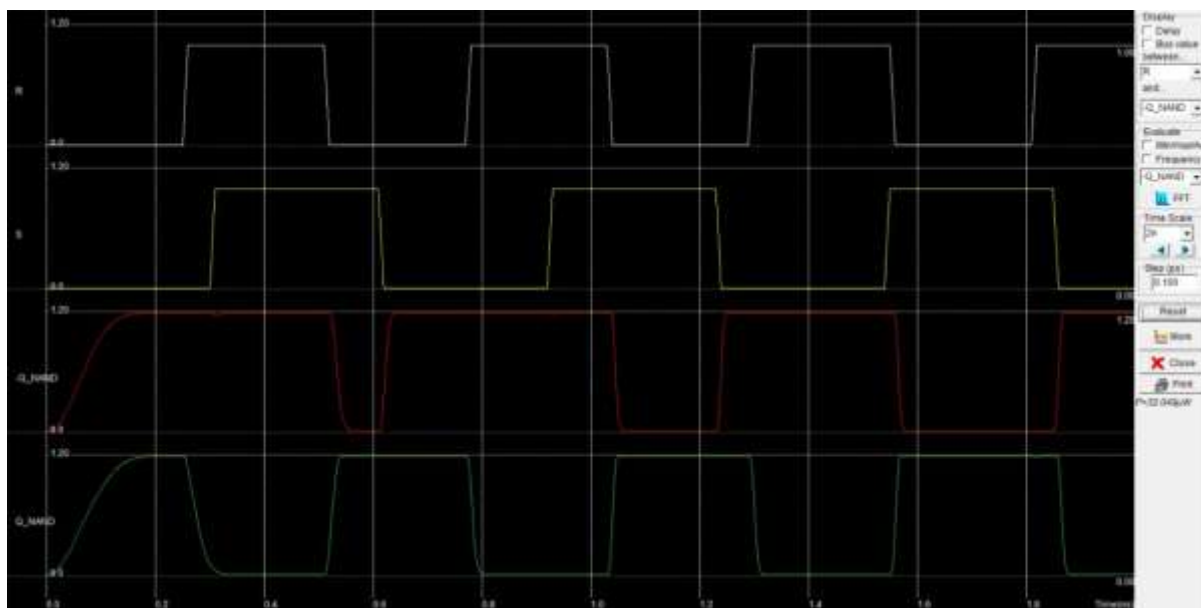
Circuit:



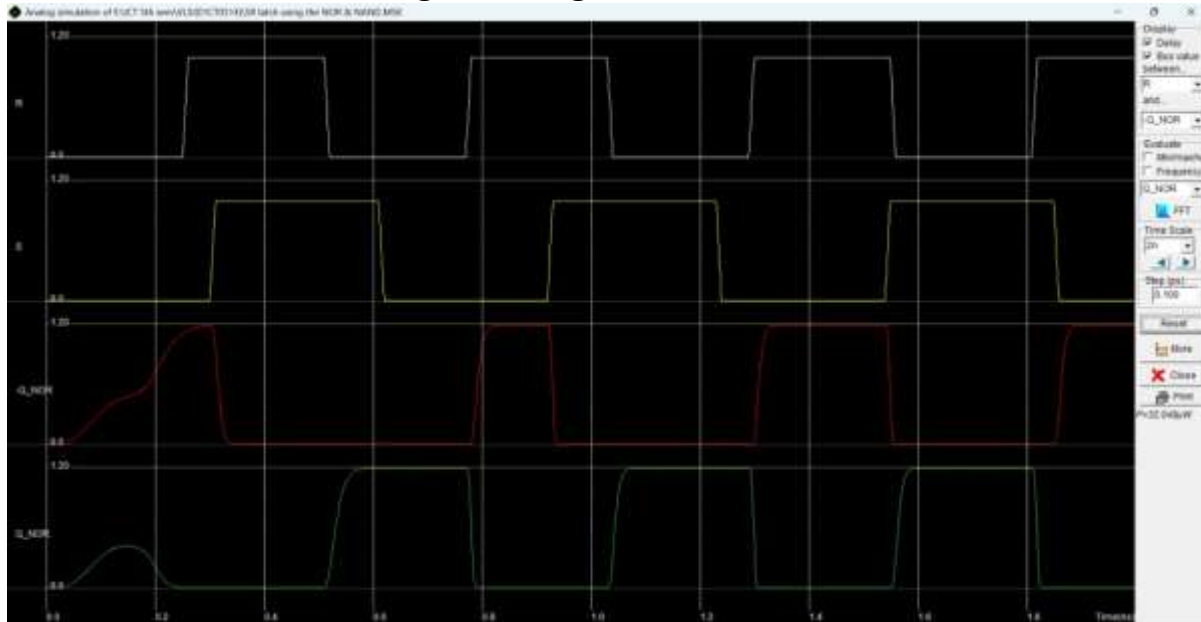


Output Waveform:-

Waveform for the SR latch using the NAND gates:



Waveform of the SR latch using the NOR gates



Conclusion:

In this experiment, the SR-latch was designed and simulated using CMOS logic. The waveforms verified its correct operation, showing that the latch can store and hold a logic state until reset. This confirms its bistable nature and importance in memory and sequential logic applications.